SUMS OF SQUARES IN CYCLOTOMIC INTEGRAL RINGS

Chun-Gang Ji and Da-Sheng Wei

Abstract Let $K_n = \mathbb{Q}(\zeta_n)$ be a cyclotomic field with algebaic integral ring O_n, where positive integer $n \not\equiv 2(\mod 4)$ and S_n the set of all elements $\alpha \in O_n$ which can be expressed as sums of squares in O_n. Set g_n the smallest positive integer m such that every element in S_n can be expressed as a sum of m squares in O_n. In this paper, we prove that (1) $g_n = 4$ if n is odd and the order of 2 in multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is odd; (2) $g_n = 3$ if n is odd and the order of 2 in multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is even; (3) $g_n = 3$ if $n \equiv 0(\mod 4)$.
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1. Introduction

Let K be an algebraic number field of degree n with exactly r real conjugates $K^{(1)}, \cdots, K^{(r)}$, in particular, the field K is totally real in the case $r = n$. A number α in K is called totally positive, $\alpha \gg 0$, whenever the r conjugates $\alpha^{(1)}$, $\cdots$, $\alpha^{(r)}$ are all positive. In 1902, Hilbert conjectured that every totally positive α in K is a sum of four squares in K. The first published proof of this was given by Siegel [Si] in 1921. Siegel had proved the following two theorems in [Si1]

Theorem 1. *Let K be totally real and suppose that all totally positive algebraic integers are sums of integral squares in K; then K is either the rational number field $\mathbb{Q}$ or the real quadratic number field $\mathbb{Q}(\sqrt{5})$.*

Theorem 2. *If K is not totally real, then all totally positive algebraic integers are sums of integral squares in K when and only when the discriminant of K is odd.*

These two results are from different aspects of quadratic forms theory: Theorem 1 deals with definite forms, and theorem 2 with indefinite forms. While the proof of theorem 1 was elegant, albeit surprisingly elementary, Siegel resorted to a generalization of the circle method to show that five integral squares applies for theorem 2, and expressed his belief that perhaps four may be possible. Using the modern powerful machinery of spinor genus, theorem 2 is easily reducible to a local question. In fact, the conjectured value of four integral squares quickly falls out.

Theorem 2^***.** *If K is not totally real, then all totally positive algebraic integers are sums of four integral squares in K when and only when the discriminant of K is odd.*

In Theorem 1, over $\mathbb{Q}$ it is well-known that all positive integer v are sums of four squares (Lagrange) and such a v a sum of three squares if and only if $v \neq 4^a(8b+7)$ (Legendre). Maass later showed via function-theoretic means that actually three squares works for $\mathbb{Q}(\sqrt{5})$, see [Ma].

In Theorem 2, when K is an imaginary quadratic number field, using some results from the algebraic K-theory of integral quadratic forms and the theory of spinor genus of quadratic forms, Estes and Hsia [EsH], [EsH1] proved that

Theorem 3. *Every algebraic integer in $K = \mathbb{Q}(\sqrt{-D})$, D a positive square free integer, can be expressed as a sum of three integral squares when and only when $D \equiv 3(\mod 8)$ and D does not admit a positive proper factorization $D = d_1 d_2$ (i.e., $d_i > 1$) which satisfies the conditions: (a) $d_1 \equiv 5, 7(\mod 8)$ and (b) $(d_2/d_1) = 1$, where $(*/*)$ is Jacobi symbol.*

Let $K_n = \mathbb{Q}(\zeta_n)$ be a cyclotomic number field with algebraic integral ring $O_n = \mathbb{Z}[\zeta_n]$, where ζ_n is a primitive n-th root of unity. If $n \equiv 2(\mod 4)$ then $K_n = \mathbb{Q}(\zeta_{n/2}) = K_{n/2}$. Hence in this note we assume that $n \not\equiv 2(\mod 4)$. Let S_n be the set of all element $\alpha \in O_n$ which can be expressed as sums of squares in O_n and set

$$g_n = \min\{m : \text{ any element in } S_n \text{ is a sum of } m \text{ integral squares } \}.$$

How to determine S_n and g_n? It is easy to see that $-1 \in S_n$ and S_n is a subring of O_n, in particular, $S_n = O_n$ if n is odd. In [Ji], we proved that every algebraic integer in O_n can be expressed as a sum of three integral squares if and only if n is odd and the order of 2 in multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is even. In this note, we shall prove that:

Theorem 4. *Let positive integer $n \not\equiv 2(\mod 4)$. Then*

$$g_n = \begin{cases} 4 & \text{if } n \text{ is odd and the order of } 2 \text{ in } (\mathbb{Z}/n\mathbb{Z})^* \text{ is odd,} \\ 3 & \text{if } n \text{ is odd and the order of } 2 \text{ in } (\mathbb{Z}/n\mathbb{Z})^* \text{ is even,} \\ 3 & \text{if } n \equiv 0(\mod 4). \end{cases}$$

2. Some Lemmas

Lemma 1. *Let K, E be two number fields with integral rings O_K and O_E respectively. Assume that their discriminants $d(K)$ and $d(E)$ are relatively prime and that the fields are linearly disjoint (i.e. if $\omega_1, \cdots, \omega_n$ is a basis of K over $\mathbb{Q}$ and $\upsilon_1, \cdots, \upsilon_m$ is a basis of E over $\mathbb{Q}$, then $\{\omega_i\upsilon_j\}$is a basis of KE over $\mathbb{Q}$). Then*

$$O_{KE} = O_K O_E \quad \text{and} \quad d(KE) = d(K)^m d(E)^n,$$

where O_{KE} is the integral ring of KE and $d(KE)$ is the discriminant of KE.

Proof. See page 68 Proposition 17 in [La].

Using Lemma 1 in cyclotomic number fields, we have $O_{mn} = O_m O_n$, where m and n are relatively prime. In particular, we have

Lemma 2. *Let $n > 2$ be a positive integer such that $n \not\equiv 2(\mod 4)$. Then*

$$d(K_n) = (-1)^{\phi(n)/2} \frac{n^{\phi(n)}}{\prod_{p|n} p^{\phi(n)/(p-1)}}.$$

Now it is not difficult to determine S_n. We have the following result

Lemma 3. *Let $n = 2^r m$, where $r = 0$ or ≥ 2 and m is odd. Then*

$$S_{2^r} = \{\alpha | \alpha = a_0 + 2a_1\zeta_{2^r} + a_2\zeta_{2^r}^2 + 2a_3\zeta_{2^r}^3 + \cdots + 2a_{2^{r-1}-1}\zeta_{2^r}^{2^{r-1}-1}, \quad a_i \in \mathbb{Z}\}$$

and

$$S_n = \{\alpha | \alpha = a_0 + 2a_1\zeta_{2^r} + a_2\zeta_{2^r}^2 + 2a_3\zeta_{2^r}^3 + \cdots + 2a_{2^{r-1}-1}\zeta_{2^r}^{2^{r-1}-1}, \quad a_i \in \mathbb{Z}[\zeta_m]\}.$$

Lemma 4. *Let $m \geq 3$ be an odd positive integer and $K = \mathbb{Q}(\zeta_m)$. Then $s(K)$ (the stufe of K, that is to say, the smallest number of squares necessary to represent -1 in K) is equal to 2 or to 4 depending on whether the order of 2 modulo m is even or odd.*

Proof. See [Mo] or [Co].

Lemma 5. *Let p be an odd prime. If the order of 2 modulo p is even, then -1 can be represented as a sum of two integral squares in $K_p = \mathbb{Q}(\zeta_p)$.*

Proof. Let $f = 2n$, $n \geq 1$, be the order of 2 modulo p. Then we have

$$2^{2n} = 2^f \equiv 1(\mod p), \quad \text{and} \quad 2^n \equiv -1(\mod p).$$

From

$$(1+\zeta_p^2)(1+\zeta_p^{2^2})(1+\zeta_p^{2^3})\cdots(1+\zeta_p^{2^n}) = \frac{1-\zeta_p^{2^{n+1}}}{1-\zeta_p^2} = \frac{1-\zeta_p^{-2}}{1-\zeta_p^2} = \frac{-1}{\zeta_p^2},$$

we have

$$-1 = \zeta_p^2(1+\zeta_p^2)(1+\zeta_p^{2^2})\cdots(1+\zeta_p^{2^n}). \tag{1}$$

Then the result follows from (1) and the following identity

$$(a^2+b^2)(c^2+d^2) = (ac+bd)^2 + (ad-bc)^2. \tag{2}$$

Lemma 6. *Let $n > 2$ be a positive integer with $n \equiv 2(\mod 4)$. If -1 can be expressed as a sum of two integral squares in $K_n = \mathbb{Q}(\zeta_n)$, then every element in S_n can be expressed as a sum of three integral squares in K_n.*

Proof. Let $\alpha \in S_n$. Then we have

$$-\alpha = \beta_1^2 + \beta_2^2 + \cdots + \beta_t^2, \quad \beta_i \in \mathbb{Z}[\zeta_n].$$

Then there exists a $\gamma \in \mathbb{Z}[\zeta_n]$ such that

$$\alpha + (\beta_1 + \beta_2 + \cdots + \beta_t + 1)^2 = (\gamma+1)^2 - \gamma^2.$$

So there exist $x, y, z \in \mathbb{Z}[\zeta_m]$ such that

$$\alpha = x^2 - (y^2 + z^2).$$

Because -1 can be expressed as a sum of two integral squares in K_n and using (2) we can obtain the result.

Lemma 7. *Let K/k be a finite abelian extension, where k is a finite extension of $\mathbb{Q}_p$. There is a map (called the Artin map)*

$$k^* \to Gal(K/k)$$

$$a \mapsto (a, K/k)$$

which induces an isomorphism

$$k^*/N_{K/k}K^* \simeq Gal(K/k),$$

where $N_{K/k}$ denotes the norm mapping. Let T denote the inertia subgroup of $Gal(K/k)$. Then

$$U(k)/N_{K/k}U(K) \simeq T,$$

where $U(k)$ and $U(K)$ are local units of k and K respectively.

Proof. See page 402 Theorem 7 in [Wa].

Lemma 8 (Local Square Theorem). *Let α be an integer in the local field F with the uniformizer π. Then there is an integer β such that*

$$1 + 4\pi\alpha = (1 + 2\pi\beta)^2.$$

Proof. See page 159, 63:1 in [O].

3. Proof of Theorem 4

Case A, suppose that n is odd and the order of 2 in multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is odd. By Lemma 2 and Theorem 2*, we have $g_n \leq 4$. By Lemma 4 and $-1 \in S_n$, we have $g_n > 3$. Hence $g_n = 4$.

Case B, suppose that n is odd and the order of 2 in multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is even. Let

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_t^{\alpha_t},$$

where the $p_j's$ are distinct odd primes. Then there exists a prime p_j $(1 \leq j \leq t)$ such that the order of 2 in $(\mathbb{Z}/p_j\mathbb{Z})^*$ is even. If the order of 2 in $(\mathbb{Z}/p_j\mathbb{Z})^*$ is odd for any $1 \leq j \leq t$, then the residue class degree above 2 in K_{p_j} is odd. using

$$\mathbb{Q} \subset K_{p_j} \subset K_{p_j^{\alpha_j}}$$

and $[K_{p_j^{\alpha_j}} : K_{p_j}] = p_j^{\alpha_j - 1}$ is odd, we have the residue class degree above 2 in $K_{p_j^{\alpha_j}}$ is odd for any $1 \leq j \leq t$. Since K_n is the composite of the fields $K_{p_j^{\alpha_j}}$ $(j = 1, 2, \cdots, t)$, we have the residue class degree above 2 in K_n is odd, this contradicts the order of 2 in $(\mathbb{Z}/n\mathbb{Z})^*$ is even. By lemmas 5 we have -1 can be expressed as a sum of two integral squares in K_n. Using Lemma 6 we have $g_n \leq 3$.

We claim that there is a unit ε in K_n such that ε can not be expressed as a sum of two integral squares in K_n. If not so , every unit in O_n is a sum of two squares in O_n. Let $\mathfrak{p}$ be a prime over 2 in K_n and consider $(K_n)_\mathfrak{p}$ the completion of K_n at $\mathfrak{p}$. Since the unit group of K_n is dense in the unit group of $(K_n)_\mathfrak{p}$. By Lemma 8, Local Square Theorem, every unit of $(K_n)_\mathfrak{p}$ is a sum of two integral squares in $(K_n)_\mathfrak{p}$. Let $L = K_n(\sqrt{-1}) = \mathbb{Q}(\zeta_{4n})$ and $\mathfrak{q}$ a prime over $\mathfrak{p}$ in L. Then $[L : K_n] = 2$ and $\mathfrak{p}$ is totally ramified in L. Let $L_\mathfrak{q}$ be the completion of L at $\mathfrak{q}$. By lemma 7 we have $|U((K_n)_\mathfrak{p})/N_{(K_n)_\mathfrak{p}/L_\mathfrak{q}} U(L_\mathfrak{q})| = 2$, that is, there is not every unit in $(K_n)_\mathfrak{p}$ is a sum of two integral squares $(K_n)_\mathfrak{p}$. Which is a contradiction. So $g_n = 3$.

Case C, suppose that $n \equiv 0(\mod 4)$. We have $\sqrt{-1} \in K_n$, so -1 can be expressed as a sum of two integral squares in K_n. By Lemma 6 we have $g_n \leq 3$. Let $n = 2^r m$, where m is odd. Then $2(1 - \zeta_{2^r})$ is not expressed as a sum of two integral squares in K_n. Hence $g_n > 2$, we have $g_n = 3$.

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Department of Mathematics
Nanjing Normal University
Nanjing 210097
China
e-mail: cgji@njnu.edu.cn

Department of Mathematics
The University of Science and Technology of China
Hefei 230026
China
e-mail: dswei@mail.ustc.edu.cn